

Full Name

Section

Hardness Test Report Template

In all questions: You can use Excel, Matlab, Python, or anything else you are comfortable with. Your report should be approximately this long; therefore, be concise in your explanations.

Metals

(a) Using the collected data from all groups (*on Moodle*) calculate the mean and the standard deviation for each specimen. Comment on possible reasons for the variation in measurement results.

(b) Write the data of your group (d_1 , d_2 , HV for each specimen) Based on d_1 and d_2 for one of the measurements, check if the measurement agrees with the equation for HV.

Copper			Steel			Aluminum		
d_1	d_2	HV	d_1	d_2	HV	d_1	d_2	HV

(c) Based on a literature search, determine which type of steel specimen was possibly employed.

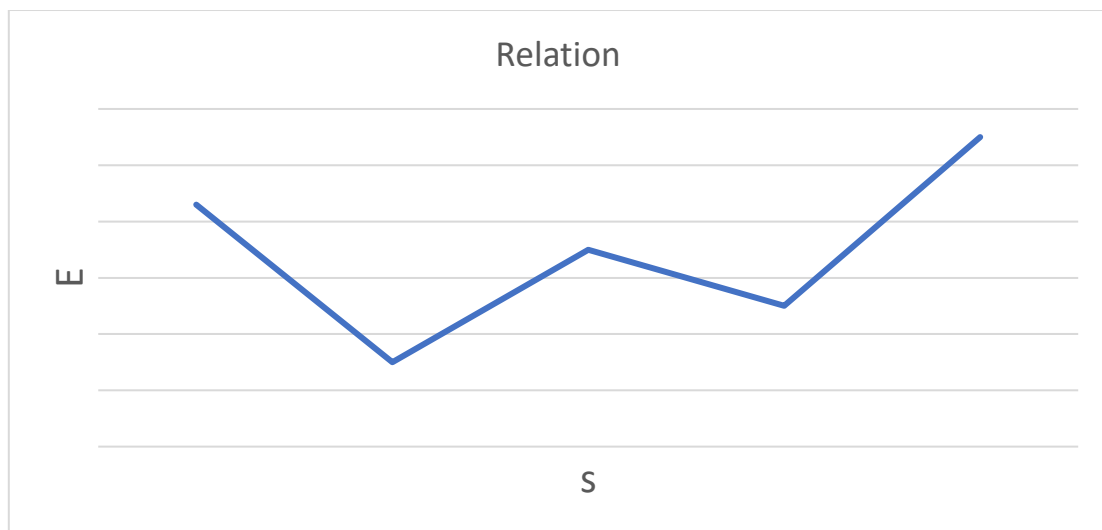
(d) Another common indentation-based hardness test is the so-called *Brinell* test. Using an appropriate conversion table, calculate the mean *Brinell Hardness* (HB) for the test specimen. An empirical relation allows one to relate the measured *HB* value to the ultimate strength of a steel specimen: $\sigma_u = 3.45 \times HB$. Calculate the ultimate strength σ_u of the test specimen. Is this consistent with your guess in part (c)?

(e) When would you desire a metal with a high or low hardness? Also, make an online search on how to manufacture a mechanical part which has high hardness on the outside but low hardness on the inside and why it would be desirable to manufacture such a part.

Polymers

(a) Using the collected data from all groups (*on Moodle*) calculate the mean and the standard deviation for each specimen. Comment on possible reasons for the variation in the measurement results as well as on the reliability of the results.

(b) Convert the mean Shore A hardness readings to Young's moduli values using the relation provided earlier. Also, plot this relation and answer the following question: do small errors in hardness readings correspond to larger or smaller errors in Young's moduli values with increasing hardness?



(c) Make an online search for typical Shore A hardness values and which materials they possibly correspond to. Do these general observations correlate with your findings from part (b)?

(d) List three reasons why the Young's modulus determined via the relation provided earlier may not match the one from a tension test.

(e) Evaluate $E(0)$ and subsequently carry out an online search to find at least two materials which cannot be characterized faithfully through the Shore A hardness test.

References

Appendix

Include your code & extra figures